

# Multi-Scale Convolutional Attention Network for Predicting Remaining Useful Life of Machinery

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**Abstract**—To integrate the complete degradation information of machinery, deep learning-based prognostics approaches usually use monitoring data acquired by different sensors as the inputs of networks. These approaches, however, lack an explicit learning mechanism to effectively identify the distinctions of different sensor data and highlight the important degradation information, thereby affecting the accuracy of deep prognostics networks and limiting their generalization. To overcome the aforementioned weaknesses, a new deep prognostics framework named multi-scale convolutional attention network (MSCAN) is proposed in this paper for predicting remaining useful life (RUL) of machinery. In the proposed MSCAN, self-attention modules are first constructed to effectively fuse the input multi-sensor data. Then, a multi-scale learning strategy is developed to automatically learn representations from different temporal scales. Finally, the learned high-level representations are fed into dynamic dense layers to perform regression analysis and RUL estimation. The proposed MSCAN is evaluated using multi-sensor monitoring data from life testing of milling cutters, and also compared with some state-of-the-art prognostics approaches. Experimental results demonstrate the effectiveness and superiority of the proposed MSCAN in fusing multi-sensor information and improving RUL prediction accuracy.

**Index Terms**—Deep learning, convolutional neural network, self-attention mechanism, multi-scale learning, remaining useful life prediction.

## I. INTRODUCTION

NOWADAYS, industrial internet of things has been widely used in condition monitoring of machinery, and a mass of

monitoring data are acquired by various sensors, such as accelerometers, dynamometers and acoustic emission (AE) sensors. The explosive growth of monitoring data brings new opportunities and challenges to remaining useful life (RUL) prediction of machinery. Since data-driven RUL prediction methods are able to learn degradation characteristics of machinery from massive monitoring data and build the corresponding RUL prediction models automatically, they are gaining more and more attention in the field of RUL prediction of machinery recently [1].

Data-driven prognostics approaches usually contain three steps: hand-crafted feature design, degradation behavior learning and RUL estimation [2]. Hand-crafted feature design is to use prior knowledge and expertise to extract sensitive degradation features from the acquired monitoring data. Then, these features are fed into machine learning models, e.g., random forests (RFs), Bayesian networks and support vector machines (SVMs), to learn the degradation behavior of machinery and estimate the RUL. For instance, Soualhi *et al.* [3] extracted sensitive features from vibration signals using Hilbert-Huang transform and these features were input into SVMs to predict RUL of rolling element bearings. Drouillet *et al.* [4] used spindle power data to train an artificial neural network and then employed it to estimate RUL of milling cutters. Cheng *et al.* [5] proposed a locally linear embedding-based feature fusion strategy to obtain a sensitive bearing degradation feature and utilized an adaptive neuro-fuzzy inference system to perform RUL estimation tasks. Although these data-driven prognostics approaches are able to infer correlations and causalities hidden in data [6], they still need to take much effort on hand-crafted feature design [7, 8]. To change this situation, it is desirable to use advanced artificial intelligence techniques to directly learn the degradation behavior of machinery from raw monitoring data. Therefore, deep learning is introduced into data-driven RUL prediction [9–11].

Deep learning is a particular kind of machine learning that is structured by a stack of multiple layers of nonlinear processing units [12]. Compared with traditional machine learning techniques, deep learning techniques, such as deep belief networks (DBNs), recurrent neural networks (RNNs) and convolutional neural networks (CNNs), have more powerful representation learning ability. They are able to learn complex functions mapping the input to the output directly from raw data, without depending completely on hand-crafted features. Deep learning has delivered state-of-the-art accuracy in many tasks, e.g., image classification, speech recognition, natural language processing, machine translation and so on. In

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data-driven RUL prediction of machinery, some deep learning-based prognostics approaches have been developed in recent years and achieved promising RUL prediction results. Babu *et al.* [9] presented a CNN-based prognostics approach for RUL prediction of turbofan engines and demonstrated its superiority by comparing it with the traditional machine learning methods. Hinch *et al.* [13] used CNNs and long short-term memory (LSTM) networks to predict RUL of rolling element bearings, in which CNNs were first used to extract local features from vibration signals and then LSTM networks were employed to perform RUL prediction. Zhang *et al.* [6] proposed a multi-objective DBN ensemble and utilized it to estimate RUL of turbofan engines. Zhu *et al.* [14] combined wavelet transform and CNNs to predict the bearing RUL. Sun *et al.* [15] constructed a deep sparse auto-encoder network and applied it into RUL prediction of cutting tools.

In existing deep learning-based prognostics approaches, monitoring data from different sensors are usually used as the inputs of deep prognostics networks so as to integrate the complete degradation information of a monitored machine. These approaches, however, lack an explicit learning mechanism to effectively fuse these multi-sensor data. During network construction, they assume that the input data acquired by different sensors make an equal contribution to the outputs. But in real cases, monitoring data from different sensors contain different degrees of degradation information. Some of them may be informative, while others may be not. If there is no mechanism to identify the distinctions of different sensor data and highlight the important degradation information, the prediction performance of deep networks will be affected by the irrelevant or redundant information, leading to poor accuracy and generalization in RUL prediction of machinery.

To overcome the aforementioned weaknesses, a new deep prognostics framework named multi-scale convolutional attention network (MSCAN) is proposed for predicting RUL of machinery in this paper. In the proposed MSCAN, self-attention modules are first constructed to effectively fuse the input multi-sensor data. Then, a multi-scale learning strategy is developed to automatically learn representations from different temporal scales. Finally, the learned high-level representations are fed into dynamic dense layers (DDLs) to perform regression analysis and RUL estimation. The proposed MSCAN is evaluated using multi-sensor monitoring data from life testing of milling cutters, and also compared with some state-of-the-art prognostics approaches. Experimental results demonstrate the effectiveness and superiority of the proposed MSCAN in fusing multi-sensor information and improving RUL prediction accuracy. The main contributions of this paper are summarized as follows:

- 1) Two self-attention mechanisms, i.e., channel-wise attention and temporal attention, are presented to adaptively identify the multi-sensor degradation information. The embedding of these two attention mechanisms enables the prognostics network to dynamically emphasize the discriminative information while suppressing the useless information, thereby facilitating the information flow within the network and enhancing the representational power of the network.

- 2) A multi-scale learning strategy is developed by using parallel convolutional pathways, which enables the prognostics network to automatically extract degradation features from different temporal scales and thus ensures the completeness of representations.

- 3) A new layer, i.e., DDL, is built to replace the traditional fully-connected layer (FCL). In the DDL, dynamic Gaussian dropout firstly offers a time-varying strategy to prevent co-adaptations between neurons, which is able to effectively relieve the over-fitting problem and enhance the generalization of the prognostics network. Then, the DDL introduces maxout activation to boost the reasoning ability, further improving the RUL prediction accuracy.

The remaining parts of this paper are organized as follows. Section II details the proposed MSCAN for RUL prediction of machinery. In Section III, multi-sensor monitoring data from life testing of milling cutters are used to demonstrate the effectiveness and superiority of the proposed MSCAN. Finally, Section IV concludes this paper.

## II. PROPOSED MSCAN FOR RUL PREDICTION OF MACHINERY

The architecture of the proposed MSCAN is illustrated in Fig. 1, which is comprised of representation learning subnetwork and RUL estimation subnetwork. In MSCAN, monitoring data acquired by different sensors are used as the inputs to integrate the multi-sensor degradation information. Then, the representation learning subnetwork is constructed to automatically discover and highlight the discriminative information from the input sensor data by embedding self-attention mechanisms and using multi-scale learning. Finally, the learned representations are fed into the RUL estimation subnetwork to perform RUL prediction. The details of MSCAN are described as follows.

### A. Representation learning subnetwork

The representation learning subnetwork is structured by a stack of multiple convolutional building blocks. In particular, to effectively fuse the degradation information from different monitoring sensors, a new module, called self-attention module, is constructed behind each convolutional building block. Further, to ensure the completeness of representations, a multi-scale learning strategy is proposed in this subnetwork.

#### 1) Convolutional building blocks

In the proposed MSCAN, each convolutional building block consists of two convolutional layers and a pooling layer. The convolutional layers aim to learn multiple levels of representations from the input sensor data. For each convolutional layer, it firstly uses a set of learnable kernels to convolve the inputs, and then applies an element-wise nonlinear activation function on the outputs of convolution operations. By these two operations, different feature maps can be obtained in a convolutional layer. Formally, let  $\mathbf{x}^{l-1} \in \mathbb{R}^{H \times l \times C}$  and  $\mathbf{k}^l \in \mathbb{R}^{F \times l \times C \times N}$  respectively denote the input volume and the learnable convolution kernels, where  $H$  is the length of the input volume,  $C$  is the number of input channels,  $F \times 1$  is the size of convolution kernels, and  $N$  is the number of convolution kernels. Then, the  $n$ -th feature map of the  $l$ -th

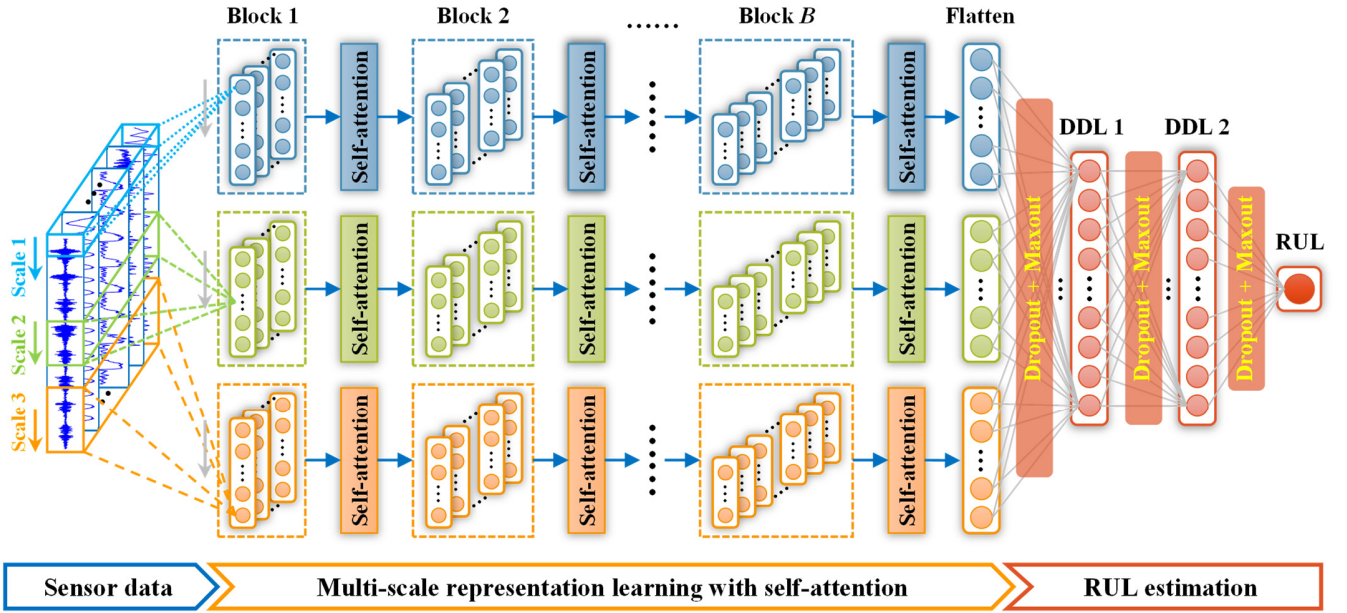


Fig. 1. Architecture of the proposed MSCAN.

convolutional layer,  $\mathbf{x}_n^l$ , can be got by

$$\mathbf{x}_n^l = \sigma_r(\mathbf{u}_n^l) \quad (1)$$

$$\mathbf{u}_n^l = \mathbf{k}_n^l * \mathbf{x}^{l-1} + \mathbf{b}_n^l = \sum_{c=1}^C \mathbf{k}_{n,c}^l * \mathbf{x}_c^{l-1} + \mathbf{b}_n^l \quad (2)$$

where  $\sigma_r(\cdot)$  is the rectified linear unit (ReLU) activation function,  $\mathbf{u}_n^l$  is the output of convolution operations,  $*$  denotes the convolution operator,  $\mathbf{k}_n^l$  is the  $n$ -th convolution kernel and  $\mathbf{b}_n^l$  is the bias term.

The use of the pooling layer is to reduce the dimensionality of representations and make them compact. In a convolutional building block, the pooling layer is placed after the second convolutional layer and conducts independently max pooling operations on every input feature map. Mathematically, the  $n$ -th feature map of the  $l$ -th pooling layer,  $\mathbf{y}_n^l$ , is able to be calculated by

$$\mathbf{y}_n^l = \text{pool}(\mathbf{y}_n^{l-1}, p, s) \quad (3)$$

where  $\mathbf{y}_n^{l-1}$  is the  $n$ -th input feature map, i.e., the  $n$ -th feature map in the previous convolutional layer,  $\text{pool}(\cdot)$  is the max pooling function,  $p$  is the pooling size, and  $s$  is the stride size. In this paper, the max pooling operations are performed using a non-overlapping window, i.e.,  $p = s$ .

## 2) Self-attention modules

The purpose of constructing self-attention modules is to effectively fuse the multi-sensor degradation information, i.e., to emphasize the important information that is relevant to the degradation states of machinery while suppressing the useless information. As shown in Fig. 1, the proposed MSCAN uses multi-channel time-series sensor data as its inputs, where each channel represents one sensor sequence. These input data have two distinct characteristics. 1) The data from different sensors (i.e., the data in different channels) contain different degrees of degradation information. In other words, some of them have

many obvious machine degradation features, but others have few ones or even only include some measurement noise. 2) The key degradation features, such as shocks caused by faults, only can be found in partial positions in one sensor sequence. Therefore, to discover and highlight the discriminative information, it is necessary to identify the distinction of different sensor data by explicitly modeling interchannel relationships along the channel dimension (CD) and to capture the pivotal degradation features by independently encoding the contextual relationships of each channel along the temporal dimension (TD). Accordingly, as shown in Fig. 2, the self-attention module is composed of two parts: channel-wise attention and temporal attention. These two attention mechanisms enable MSCAN to effectively learn “what” and “where” to attend in the channel and temporal dimensions, facilitating the information flow within the network and enhancing the representational power of the network.

Given the feature maps  $\mathbf{z}^{l-1} \in \mathbb{R}^{I \times J}$  extracted from different input sensor sequences, as illustrated in Fig. 2, the self-attention module sequentially infers channel-wise attention weights  $\mathbf{a}^l \in \mathbb{R}^{1 \times J}$  and temporal attention weights  $\mathbf{\beta}^l \in \mathbb{R}^{I \times 1}$ , where  $I$  is the length of each feature map,  $J = N \times S$  is the number of the feature maps and  $S$  is the channel number of the input sensor sequences. Correspondingly, the complete attention process can be summarized as

$$\tilde{\mathbf{z}}^l = \mathbf{a}^l \otimes \mathbf{z}^{l-1} = \Phi_c(\mathbf{z}^{l-1}) \otimes \mathbf{z}^{l-1} \quad (4)$$

$$\mathbf{z}^l = \mathbf{\beta}^l \otimes \tilde{\mathbf{z}}^l = \Phi_t(\tilde{\mathbf{z}}^l) \otimes \tilde{\mathbf{z}}^l \quad (5)$$

where  $\otimes$  denotes the element-wise multiplication,  $\tilde{\mathbf{z}}^l \in \mathbb{R}^{I \times J}$  is the channel-refined feature maps, i.e., the outputs of the channel-wise attention,  $\mathbf{z}^l \in \mathbb{R}^{I \times J}$  is the final refined output feature maps, i.e., the outputs of the temporal attention, and  $\Phi_c(\cdot)$  and  $\Phi_t(\cdot)$  are respectively the channel-wise and

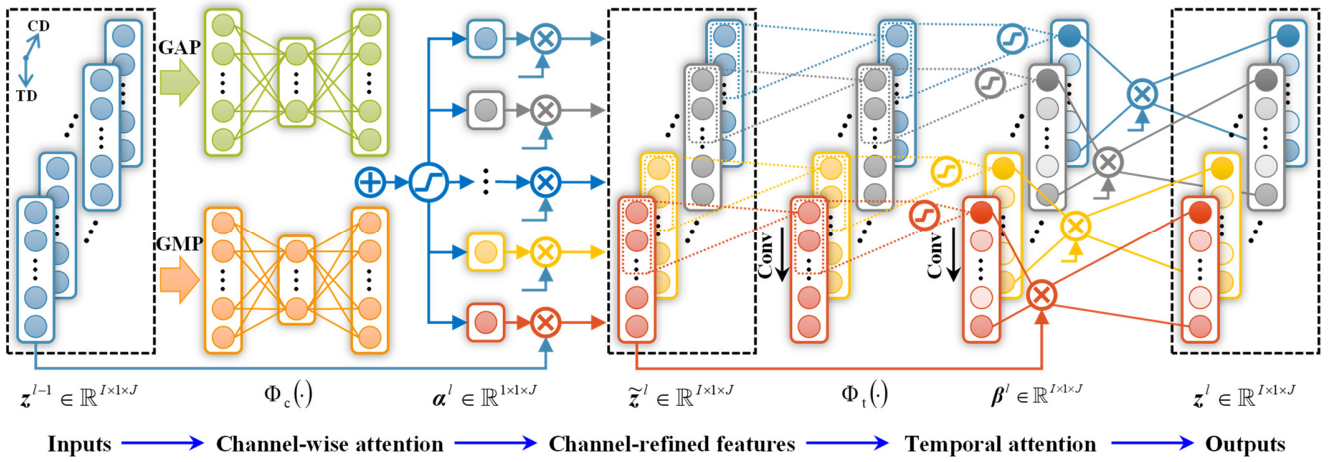


Fig. 2. An illustration for self-attention modules in MSCAN.

temporal attention functions, which are described in detail as follows.

a) *Channel-wise attention*: The channel-wise attention is built by modeling the interrelationships between channels. To achieve this, the global average pooling (GAP) and the global max pooling (GMP) are firstly used to aggregate the global information of each channel, generating two different channel descriptors  $\mathbf{v}^l \in \mathbb{R}^J$  and  $\mathbf{m}^l \in \mathbb{R}^J$ . Here,  $\mathbf{v}^l$  and  $\mathbf{m}^l$  separately contain  $J$  channel-wise statistics that are obtained by the following equations:

$$\mathbf{v}_j^l = \frac{1}{I} \sum_{i=1}^I z_{j,i}^{l-1} \quad (6)$$

$$\mathbf{m}_j^l = \max(z_j^{l-1}). \quad (7)$$

Then,  $\mathbf{v}^l$  and  $\mathbf{m}^l$  are forwarded to a multi-layer perceptron (MLP) with one hidden layer to capture the interchannel relationships and estimate the informativeness of every channel, respectively. Especially, to reduce the computational complexity, the neuron number of the hidden layer in the MLPs is set to  $J/r$ , where  $r$  is the ratio of dimensionality reduction. After that, the outputs of two MLPs are merged by using element-wise summation and the channel-wise attention weights  $\alpha^l$  are got by

$$\alpha^l = \sigma_{h_s}(\mathbf{W}_{12}^l(\mathbf{W}_{11}^l \mathbf{v}^l) \oplus \mathbf{W}_{22}^l(\mathbf{W}_{21}^l \mathbf{m}^l)) \quad (8)$$

where  $\oplus$  denotes the element-wise summation,  $\sigma_{h_s}(\cdot)$  is the

hard sigmoid activation function [16],  $\mathbf{W}_{11}^l \in \mathbb{R}^{\frac{J}{r} \times \frac{J}{r}}$ ,

$\mathbf{W}_{12}^l \in \mathbb{R}^{\frac{J}{r} \times \frac{J}{r}}$ ,  $\mathbf{W}_{21}^l \in \mathbb{R}^{\frac{J}{r} \times J}$  and  $\mathbf{W}_{22}^l \in \mathbb{R}^{J \times \frac{J}{r}}$  are the weight matrices in the MLPs, and the biases are omitted for simplifying notations. Finally, the channel-refined feature maps  $\tilde{z}^l$  are produced by conducting channel-wise multiplication between the inputs  $z^{l-1}$  and the channel-wise attention weights  $\alpha^l$  along the channel dimension, i.e.,  $\tilde{z}_j^l = \alpha_j^l \cdot z_j^{l-1}$ . It is worth noting that each feature map obtained by the convolutional building block acts as a pattern detector [17]. During representation learning, these detectors are able to automatically extract various patterns from the multi-sensor input data, including informative patterns and useless ones.

Hence, the channel-wise attention can be viewed as a process of recognizing important degradation patterns. That is, the channel-wise attention focuses on “what” is useful given the multi-sensor data.

b) *Temporal attention*: In one input sensor sequence, there may be some distinctive degradation features, e.g. periodic shocks or impulses. These features play key roles in RUL prediction of machinery. However, it should be noted that the distribution of these features is discrete along the time axis and thus they merely can be found in some local locations of the sensor sequence. Correspondingly, if the information extracted from the input sensor sequence is directly used to predict the RUL, it may lead to sub-optimal results due to the irrelevant locations. In consequence, instead of considering each temporal location equally, the temporal attention attempts to pay more attention to the locations that are relative to machine degradation. That is, the temporal attention focuses on “where” is salient given one sensor sequence, which is complementary to channel-wise attention.

The temporal attention captures informative locations by encoding the contextual relationships of each channel. As shown in Fig. 2, in the temporal attention, depthwise dilated convolutions are first employed to convolve the channel-refined feature maps  $\tilde{z}^l$ . Different from standard convolutions, the depthwise dilated convolutions apply a single dilated convolution kernel into each channel of  $\tilde{z}^l$  and so can independently map the context of each channel. Meanwhile, the use of dilated convolutions [18] is able to enlarge the receptive field while keeping efficient computation. After two depthwise dilated convolutional layers, the hard sigmoid function is adopted to implement the nonlinear activation, resulting in the temporal attention weights  $\beta^l$ . Mathematically, the  $j$ -th weight vector of  $\beta^l$  can be computed by

$$\beta_j^l = \sigma_{h_s}(\mathbf{D}_{2,j}^l * (\mathbf{D}_{1,j}^l * \tilde{z}_j^l + b_{1,j}^l) + b_{2,j}^l) \quad (9)$$

where  $*_d$  denotes the dilated convolution operation with a dilation rate of size  $d$ ,  $\mathbf{D}_{1,j}^l \in \mathbb{R}^{F' \times 1}$  and  $\mathbf{D}_{2,j}^l \in \mathbb{R}^{F' \times 1}$  are the dilated convolution kernels corresponding to the  $j$ -th channel,  $F' \times 1$  is the size of dilated convolution kernels,  $b_{1,j}^l$  and  $b_{2,j}^l$  are the bias terms. Finally, by performing element-wise



multiplication between  $\tilde{z}^l$  and  $\beta^l$ , the refined feature maps  $z^l$  are got and fed into the subsequent convolutional building blocks.

### 3) Multi-scale representation learning via parallel convolutional pathways

In convolutional networks, it is crucial to select the size of convolution kernels. This is because the convolution kernels with different sizes are able to extract information from different temporal scales [19]. To be specific, a large convolution kernel is preferred for information that is distributed more globally, whereas a small convolution kernel is preferred for information that is distributed more locally. For the monitoring data acquired by multiple sensors, the distribution of pivotal degradation features is varying over operating time. At the initial stage of machine degradation, there are only few degradation features in the monitoring data, but as time goes on, faults become severer and accordingly more and more degradation features are included in the monitoring data. In addition, for different kinds of monitoring data, such as vibration signals and AE signals, the scales of degradation features are also diverse. As a result, if just a single convolution kernel size is used in the deep prognostics networks, it is inevitable for the loss of degradation information during representation learning, leading to the reduction of prediction accuracy. To avoid such an issue, a multi-scale learning strategy is developed in this paper.

As displayed in Fig. 1, the multi-scale learning strategy uses three different convolutional pathways to learn representations from the input sensor data. These convolutional pathways are arranged in parallel and separately utilize three different convolution kernel sizes, i.e.,  $F \times 1$ ,  $2F \times 1$  and  $4F \times 1$ . During representation learning, there is no interaction among the three convolutional pathways and the degradation information is independently extracted from different temporal scales, thereby ensuring the completeness of representations. Moreover, it should be illustrated that except for using different convolution kernel sizes, the three convolutional pathways have the same network structure and hyperparameter settings. For each convolutional pathway, the high-level representations can be learned by stacking  $B$  convolutional building blocks and  $B$  self-attention modules, and then these learned representations are flattened as a 1-dimensional vector. After that, the three obtained vectors are concatenated together and are input into the RUL estimation subnetwork to perform RUL prediction.

#### B. RUL estimation subnetwork

In the RUL estimation subnetwork, instead of the traditional FCL, a new layer, DDL, is built to perform regression analysis and reasoning. As shown in Fig. 3, there are two key techniques to be included in the DDL. One is the dynamic Gaussian dropout that is developed to relieve the over-fitting problem and enhance the generalization of the prognostics network. Different from standard dropout that randomly masks neurons with a fixed probability of Bernoulli distribution [20], the dynamic Gaussian dropout developed in this paper applies multiplicative noise sampled from  $\mathcal{N}(1, \text{var}(t))$  into each neuron of the inputs. In particular, the variance of Gaussian noise is varying over the training time so as to dynamically prevent co-adaptations between neurons. The other one is the

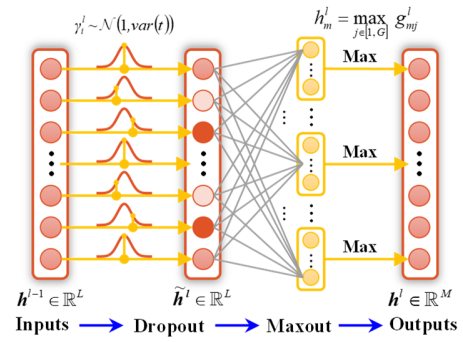


Fig. 3. Dynamic Gaussian dropout and maxout activation in the DDL.

maxout activation [21] that is employed to boost the reasoning ability of DDL and improve the prediction accuracy. The maxout activation takes the maximum of  $G$  candidate linear units as the output and is able to make a piecewise linear approximation to an arbitrary convex function.

In the proposed MSCAN, three DDLs are stacked to form the RUL estimation subnetwork. The first two DDLs have the same neuron number that is set to be  $M$ . The third DDL has only one neuron and is used as the output layer of MSCAN to predict the RUL. For the  $m$ -th neuron of the  $l$ -th DDL, its output,  $h_m^l$ , is able to be calculated by

$$\gamma_i^l \sim \mathcal{N}(1, \text{var}(t)) \quad (10)$$

$$\tilde{h}^{l-1} = \gamma^l \otimes h^{l-1} \quad (11)$$

$$g_{mj}^l = W_{mj}^l \tilde{h}^{l-1} + b_{mj}^l \quad (12)$$

$$h_m^l = \max_{j \in [1, G]} g_{mj}^l \quad (13)$$

where  $h^{l-1}$  is the inputs of the  $l$ -th DDL, e.g., the high-level representations from the representation learning subnetwork,  $\gamma_i^l$  is the Gaussian noise sampled from  $\mathcal{N}(1, \text{var}(t))$ ,  $i = 1, 2, \dots, L$ ,  $L$  is the number of the inputs,  $W_{mj}^l$  is the weight vector, and  $b_{mj}^l$  is the bias term. During training, the variance of Gaussian noise is increased linearly, i.e.,  $\text{var}(t) = t/T$ , where  $t$  is the current training epoch and  $T$  is the total number of training epochs.

## III. EXPERIMENTAL DEMONSTRATION

In this section, multi-sensor monitoring data collected from life testing of milling cutters are used to evaluate the prediction performance of the proposed MSCAN. Moreover, the RUL prediction results of MSCAN are also compared with those of four state-of-the-art prognostics approaches.

### A. Data description

As shown in Fig. 4, the life testing of milling cutters is conducted in a computer numerical control (CNC) milling machine. The material of workpieces is C45 grade steel and the type of milling cutters is APMT 1135. During machining, the table feeds workpieces from left to right along X-axis, which is referred to as a pass. As tabulated in Table I, a total of 18 milling cutters are tested under three different cutting conditions, and all tests are carried out without the application of a cutting fluid according to international standard ISO 8688–

1. To monitor failure progressions of milling cutters, as shown in Fig. 4, five types of sensors are installed in the milling machine, including triaxial accelerometer (PCB 356A15), three-component dynamometer (Kistler 9257B), free-field microphone (Brüel & Kjær 4966), current sensor (Anyway CSA201-P030T01) and AE sensor (PAC WD). For the first four sensors, the sampling frequency is set to be 10 kHz and  $1 \times 10^4$  data points are recorded every pass. For the AE sensor, the sampling frequency is 1 MHz and  $1 \times 10^6$  data points are recorded every pass.

Fig. 5 shows cutter deterioration photographs after reaching the end of life. It can be seen that the deterioration of milling cutters contains two major classes: wear and brittle fracture. For cutter wear, a GP-300C industrial microscope is used to measure the width of the flank wear land after each pass. When the width of the flank wear band is greater than 0.3mm, the tested cutter attains the deterioration limit and the life testing is stopped. Fig. 6 plots the acquired monitoring signals of C1\_1 during the whole operating life. From Fig. 6, it can be observed that some of these signals have obvious degradation trends over operating time, but others have not. In this case study, all of the acquired signals are used as the network inputs to verify the effectiveness of the proposed MSCAN in fusing multi-sensor

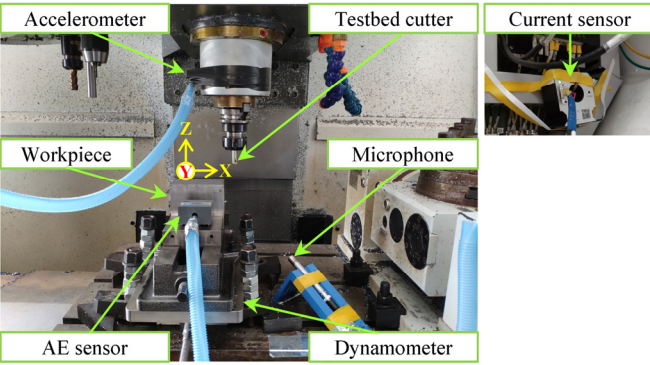


Fig. 4. CNC milling machine and sensor placement.

TABLE I  
CUTTING CONDITIONS OF MILLING CUTTERS

| Cutting condition | Spindle speed | Feed rate  | Depth of cut | Training dataset    | Testing dataset |
|-------------------|---------------|------------|--------------|---------------------|-----------------|
| Condition 1       | 2500 rpm      | 200 mm/min | 2 mm         | C1_1 C1_2 C1_3 C1_4 | C1_5 C1_6       |
| Condition 2       | 3500 rpm      | 300 mm/min | 1.75 mm      | C2_1 C2_2 C2_3 C2_4 | C2_5 C2_6       |
| Condition 3       | 4500 rpm      | 400 mm/min | 1.5 mm       | C3_1 C3_2 C3_3 C3_4 | C3_5 C3_6       |

TABLE II  
CONFIGURATION OF MSCAN IN RUL PREDICTION OF MILLING CUTTERS

| Hyperparameter                       | value | Hyperparameter                                    | value        |
|--------------------------------------|-------|---------------------------------------------------|--------------|
| Number of convolution kernels $N$    | 8     | Size of convolution kernels $F \times I$          | $4 \times 1$ |
| Pooling size $p$                     | 8     | Number of blocks $B$                              | 3            |
| Ratio of dimension reduction $r$     | 8     | Size of dilated convolution kernels $F' \times I$ | $4 \times 1$ |
| Number of candidate linear units $G$ | 4     | Dilation rate $d$                                 | 2            |
| Number of neurons $M$                | 25    | Number of training epochs $T$                     | 100          |

information and improving RUL prediction accuracy. Correspondingly, the size of an input sample is  $10000 \times 1 \times 9$ .

### B. Configuration of MSCAN

In RUL prediction of milling cutters, the hyperparameters of MSCAN are tuned and selected by performing 4-fold cross-validation on the training datasets, and the final hyperparameter selection results are listed in Table II. Particularly, these hyperparameter values are determined by considering the trade-off between prediction accuracy and computational costs. During training, this paper uses mean square error as the loss function of MSCAN and utilizes Adam optimizer with a mini-batch size of 128 to iteratively update the weights and biases of MSCAN. After 100 training epochs, the trained model is used to predict the RUL of each milling cutter in the testing datasets. Furthermore, in the following studies, each prognostics method is run 10 times for reducing the effect of randomness, and two commonly-used prognostics metrics, i.e., scoring function [22] and root mean square error (RMSE) are employed to quantitatively evaluate the prediction performance. It should be noted that the lower the score value is, the more accurate the RUL prediction results are. For each cutter in the testing datasets, the score and RMSE values are calculated from the half to the end of life.

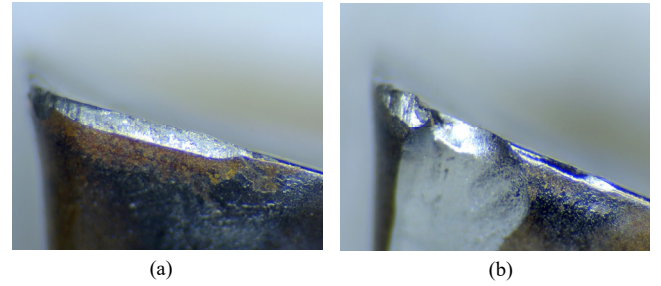


Fig. 5. Cutter deterioration photographs. (a) Flank wear. (b) Brittle fracture.

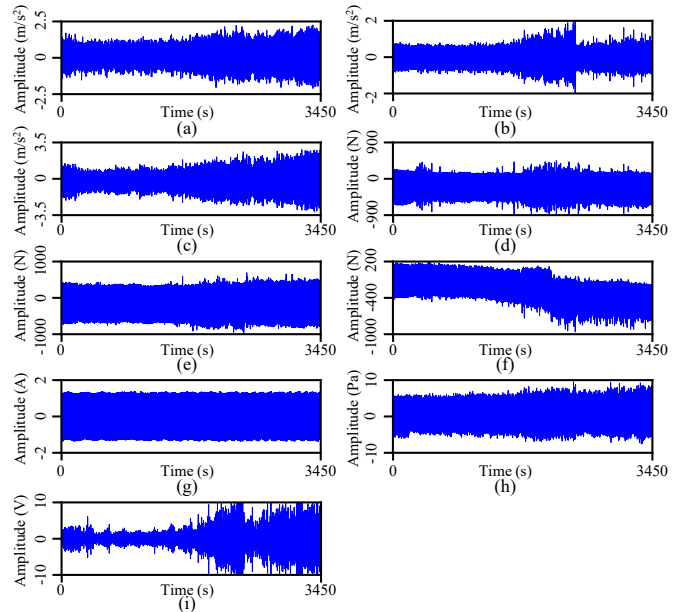


Fig. 6. Monitoring signals of C1\_1 during the whole operating life. (a) Vibration signals in X-axis. (b) Vibration signals in Y-axis. (c) Vibration signals in Z-axis. (d) Force signals in X-axis. (e) Force signals in Y-axis. (f) Force signals in Z-axis. (g) Current signals. (h) Sound signals. (i) AE signals.

### C. Benefits from self-attention mechanisms and multi-scale learning

Fig. 7 shows the RUL prediction results of C1\_5 and C1\_6, in which the predictive mean is indicated by the blue solid line, and the predictive standard deviation (SD) interval is displayed by the orange shaded region. It can be clearly seen from Fig. 7 that with the increase of cutting time, the predictive mean achieved by MSCAN converges to the actual RUL gradually, and the SD interval is also narrower. Further, to illustrate the benefits from the proposed self-attention mechanisms and multi-scale learning, the other three prognostics networks are also employed to predict the RUL and they are denoted as Network-1, Network-2 and Network-3, respectively. The architectures of these three networks are similar to that of MSCAN, and the differences are that 1) Network-1 does not use self-attention mechanisms and multi-scale learning, 2) Network-2 only uses self-attention mechanisms, and 3) Network-3 only uses multi-scale learning. In addition, the hyperparameter settings of these three networks are the same as those of MSCAN. Table III reports the performance estimation results of these four different networks. It can be observed that compared with the classic convolutional network (i.e., Network-1), the use of self-attention mechanisms or multi-scale learning effectively improves the prediction performance of the network, resulting in lower score values and smaller RMSE values. For Network-2, the performance improvement is attributed to the embedding of self-attention modules, which enables the network to adaptively identify multi-sensor degradation information along the channel and temporal dimensions. For Network-3, the employment of multiple convolutional pathways makes the network learn more complete representations, thereby improving the prediction accuracy. Besides, it is to be noted that through systematically integrating self-attention mechanisms and multi-scale learning, the proposed MSCAN obtains the lowest score value and the smallest RMSE value for each cutter among four different prognostics networks, which verifies again the benefits from self-attention mechanisms and multi-scale learning.

### D. Necessities of dynamic Gaussian dropout and maxout activation

In the proposed MSCAN, a new layer called DDL is constructed to replace the traditional FCL to perform regression analysis and reasoning. To illustrate the necessities and advantages of DDL, as shown in Fig. 8, a detailed comparison is made in this section. The compared three prognostics networks, i.e., Network-4, Network-5 and Network-6, have the same representation learning subnetwork as MSCAN, but their RUL estimation subnetworks are different. To be specific, Network-4 uses three FCLs to build the RUL estimation subnetwork, in which each FCL adopts ReLU to implement nonlinear activation and the neuron number of the first two FCLs is set to be 100. Similarly, the RUL estimation subnetworks of Network-5 and Network-6 are also comprised of three FCLs. However, different from Network-4, these two networks respectively utilize the standard dropout technique (Network-5) and the proposed dynamic Gaussian dropout technique (Network-6) during training. From Fig. 8, the first observation is that all of the networks using dropout outperform

Network-4 in terms of both the score and RMSE, which implies that it is very necessary to apply a dropout technique in the RUL estimation subnetwork. This is because, on the one hand, dropout is able to reduce the redundancy of the high-level representations from the representation learning subnetwork and prevent the over-fitting problem. On the other hand, dropout boosts the generalization ability of the prognostics networks, and accordingly, even though the training samples are limited, the networks still get accurate RUL prediction results. Especially, since the proposed dynamic Gaussian

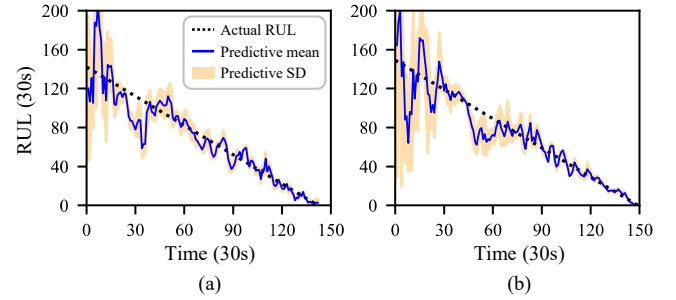


Fig. 7. RUL prediction results of milling cutters. (a) C1\_5. (b) C1\_6.

TABLE III  
PERFORMANCE ESTIMATION RESULTS OF FOUR DIFFERENT NETWORKS

| Testing dataset |       | Network-1      | Network-2     | Network-3      | MSCAN               |
|-----------------|-------|----------------|---------------|----------------|---------------------|
| C1_5            | Score | 126.83 ± 8.53  | 30.72 ± 3.36  | 44.00 ± 6.25   | <b>22.27 ± 2.91</b> |
|                 | RMSE  | 10.15 ± 0.25   | 4.73 ± 0.56   | 5.78 ± 0.82    | <b>3.82 ± 0.44</b>  |
| C1_6            | Score | 65.39 ± 11.68  | 33.90 ± 11.97 | 48.04 ± 4.09   | <b>20.61 ± 5.28</b> |
|                 | RMSE  | 7.26 ± 0.16    | 4.63 ± 0.99   | 6.09 ± 0.73    | <b>3.03 ± 0.76</b>  |
| C2_5            | Score | 58.28 ± 6.65   | 29.02 ± 6.04  | 42.51 ± 5.02   | <b>19.91 ± 2.02</b> |
|                 | RMSE  | 7.31 ± 0.31    | 4.35 ± 0.95   | 5.64 ± 0.22    | <b>3.20 ± 0.07</b>  |
| C2_6            | Score | 193.50 ± 24.69 | 111.57 ± 8.74 | 137.55 ± 12.15 | <b>70.36 ± 8.38</b> |
|                 | RMSE  | 12.55 ± 0.84   | 7.66 ± 0.99   | 9.86 ± 0.83    | <b>6.05 ± 0.32</b>  |
| C3_5            | Score | 100.13 ± 8.53  | 62.08 ± 6.30  | 74.21 ± 8.72   | <b>53.77 ± 6.10</b> |
|                 | RMSE  | 8.24 ± 0.77    | 5.44 ± 0.69   | 6.30 ± 0.49    | <b>4.85 ± 0.20</b>  |
| C3_6            | Score | 96.03 ± 7.57   | 58.03 ± 7.13  | 79.05 ± 8.18   | <b>49.92 ± 5.65</b> |
|                 | RMSE  | 9.38 ± 0.59    | 6.55 ± 0.75   | 8.01 ± 0.57    | <b>5.47 ± 0.86</b>  |

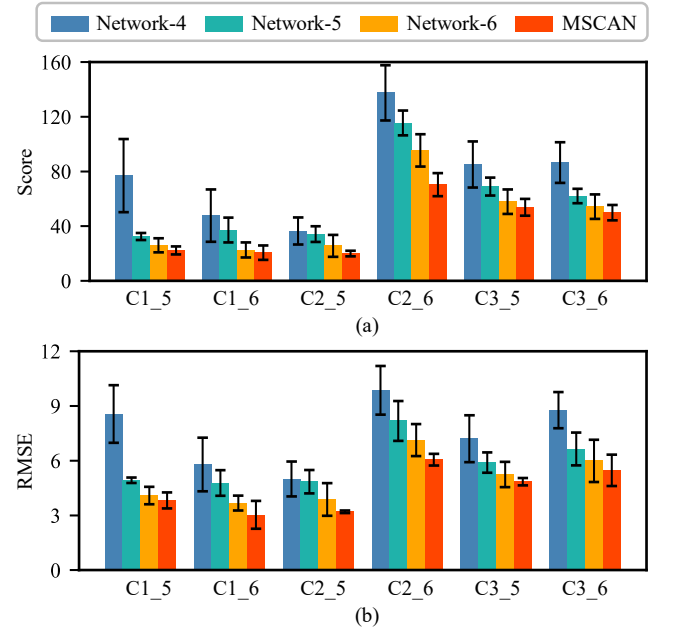


Fig. 8. Performance estimation results of four different prognostics networks. (a) Score values. (b) RMSE values.

dropout offers a time-varying co-adaptation prevention strategy, it obtains a lower score value and a smaller RMSE value for each cutter as compared to the standard dropout. As a result, the dynamic Gaussian dropout has an advantage over the standard dropout in prognostics. Moreover, it can be seen from Fig. 8 that since the proposed MSCAN introduces maxout activation to replace ReLU, it achieves higher RUL prediction accuracy than Network-6, which shows the effectiveness of maxout activation in improving the prediction performance of the network.

### E. Comparison with state-of-the-art prognostics approaches

To demonstrate the superiority of the proposed MSCAN, four state-of-the-art prognostics approaches are also employed to predict the RUL of milling cutters in this section, including RF [23], DBN [24], residual dense network (RDN) [25] and convolutional LSTM (CLSTM) [13]. For RF and DBN, features listed in [23] and [24] are first extracted from vibration, AE and force signals, and then input into the corresponding models to perform RUL prediction tasks. Whereas for RDN and CLSTM, the prognostics networks directly use all raw monitoring signals as the inputs for training and testing. Furthermore, to obtain better RUL prediction results, hyperparameter tuning is also carried out by using cross-validation for each compared approach. Table IV summarizes the performance comparison results of the proposed MSCAN and the four existing prognostics approaches in RUL prediction of milling cutters. From Table IV, it can be found that for each cutter, the proposed MSCAN has a lower score value and a smaller RMSE value than the other four prognostics approaches, which signifies that MSCAN is able to provide more accurate RUL prediction results and its performance is also more robust. Consequently, the proposed MSCAN is superior to the other four existing prognostics approaches in RUL prediction of milling cutters. This performance enhancement demonstrates again the advantages of self-attention mechanisms, multi-scale learning and DDLs.

### IV. CONCLUSION

In this paper, a new deep learning-based prognostics framework called MSCAN is proposed for predicting RUL of machinery, which is comprised of representation learning subnetwork and RUL estimation subnetwork. In MSCAN, monitoring data from different sensors are used as the inputs of the prognostics network so as to integrate the complete

degradation information. Then, the representation learning subnetwork is constructed by stacking multiple convolutional building blocks to automatically learn discriminative representations from the inputs. Particularly, behind each convolutional building block, a self-attention module is constructed to effectively fuse the multi-sensor degradation information. Meanwhile, to ensure the completeness of representations, a multi-scale learning strategy is also developed in the representation learning subnetwork. Finally, the learned representations are fed into the RUL estimation subnetwork to perform regression analysis and RUL prediction. The proposed MSCAN is experimentally validated using multi-sensor monitoring data from life testing of milling cutters, and also compared with some state-of-the-art prognostics approaches. Experimental results show that the proposed MSCAN is able to provide more accurate RUL prediction results and its prediction performance is also more robust in comparison to some existing prognostics approaches.

It should be mentioned that although the proposed MSCAN has achieved high accuracy in RUL prediction of machinery, there are two possible limitations for MSCAN in industrial applications. 1) Because of the embedding of self-attention mechanisms and the use of multi-scale learning, the total model parameters of MSCAN are more than some classic convolutional networks. Accordingly, MSCAN requires more computational costs during model training. 2) MSCAN outputs a point estimation result rather than a probabilistic distribution result in RUL prediction of machinery, and correspondingly the uncertainty of RUL prediction results cannot be obtained. Therefore, in the future research, some new methods or techniques will be developed to reduce the model size of MSCAN and quantify its prediction uncertainty.

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TABLE IV  
PERFORMANCE COMPARISON OF FIVE DIFFERENT PROGNOSTICS APPROACHES IN RUL PREDICTION OF MILLING CUTTERS

| Prognostics approach | Score (mean $\pm$ standard deviation) |                                    |                                    |                                    |                                    |                                    | RMSE (mean $\pm$ standard deviation) |                                   |                                   |                                   |                                   |                                   |
|----------------------|---------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|                      | C1_5                                  | C1_6                               | C2_5                               | C2_6                               | C3_5                               | C3_6                               | C1_5                                 | C1_6                              | C2_5                              | C2_6                              | C3_5                              | C3_6                              |
| RF [23]              | 235.82 $\pm$ 15.08                    | 145.25 $\pm$ 9.49                  | 117.89 $\pm$ 14.98                 | 302.83 $\pm$ 35.69                 | 160.28 $\pm$ 15.14                 | 128.19 $\pm$ 11.45                 | 17.32 $\pm$ 1.12                     | 9.94 $\pm$ 1.06                   | 8.48 $\pm$ 0.90                   | 16.37 $\pm$ 1.59                  | 11.01 $\pm$ 1.02                  | 12.12 $\pm$ 0.89                  |
| DBN [24]             | 146.58 $\pm$ 7.15                     | 77.06 $\pm$ 9.55                   | 50.25 $\pm$ 10.11                  | 218.19 $\pm$ 27.74                 | 100.54 $\pm$ 11.81                 | 88.45 $\pm$ 10.57                  | 13.96 $\pm$ 0.98                     | 7.42 $\pm$ 0.42                   | 6.02 $\pm$ 0.97                   | 13.39 $\pm$ 0.91                  | 8.87 $\pm$ 1.29                   | 8.17 $\pm$ 0.62                   |
| RDN [25]             | 111.65 $\pm$ 11.88                    | 53.71 $\pm$ 7.04                   | 63.08 $\pm$ 7.11                   | 168.82 $\pm$ 22.65                 | 108.66 $\pm$ 9.26                  | 103.60 $\pm$ 10.88                 | 10.79 $\pm$ 0.59                     | 6.19 $\pm$ 0.87                   | 7.64 $\pm$ 0.38                   | 11.71 $\pm$ 1.25                  | 9.01 $\pm$ 0.72                   | 9.98 $\pm$ 0.99                   |
| CLSTM [13]           | 41.47 $\pm$ 5.07                      | 35.21 $\pm$ 8.99                   | 31.74 $\pm$ 7.04                   | 120.31 $\pm$ 10.19                 | 65.60 $\pm$ 8.18                   | 61.32 $\pm$ 7.43                   | 5.10 $\pm$ 0.42                      | 4.58 $\pm$ 0.50                   | 4.63 $\pm$ 0.46                   | 8.21 $\pm$ 0.65                   | 5.62 $\pm$ 0.68                   | 7.15 $\pm$ 0.51                   |
| MSCAN                | <b>22.27 <math>\pm</math> 2.91</b>    | <b>20.61 <math>\pm</math> 5.28</b> | <b>19.91 <math>\pm</math> 2.02</b> | <b>70.36 <math>\pm</math> 8.38</b> | <b>53.77 <math>\pm</math> 6.10</b> | <b>49.92 <math>\pm</math> 5.65</b> | <b>3.82 <math>\pm</math> 0.44</b>    | <b>3.03 <math>\pm</math> 0.76</b> | <b>3.20 <math>\pm</math> 0.07</b> | <b>6.05 <math>\pm</math> 0.32</b> | <b>4.85 <math>\pm</math> 0.20</b> | <b>5.47 <math>\pm</math> 0.86</b> |



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